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Course –ECA5601- 5g/6g Communication

Experiment: 11

Implementation of IoT-based Smart Agriculture using MATLAB

Aim

To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

Objectives

1.To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.
2. To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions.
3. To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level.

Procedure

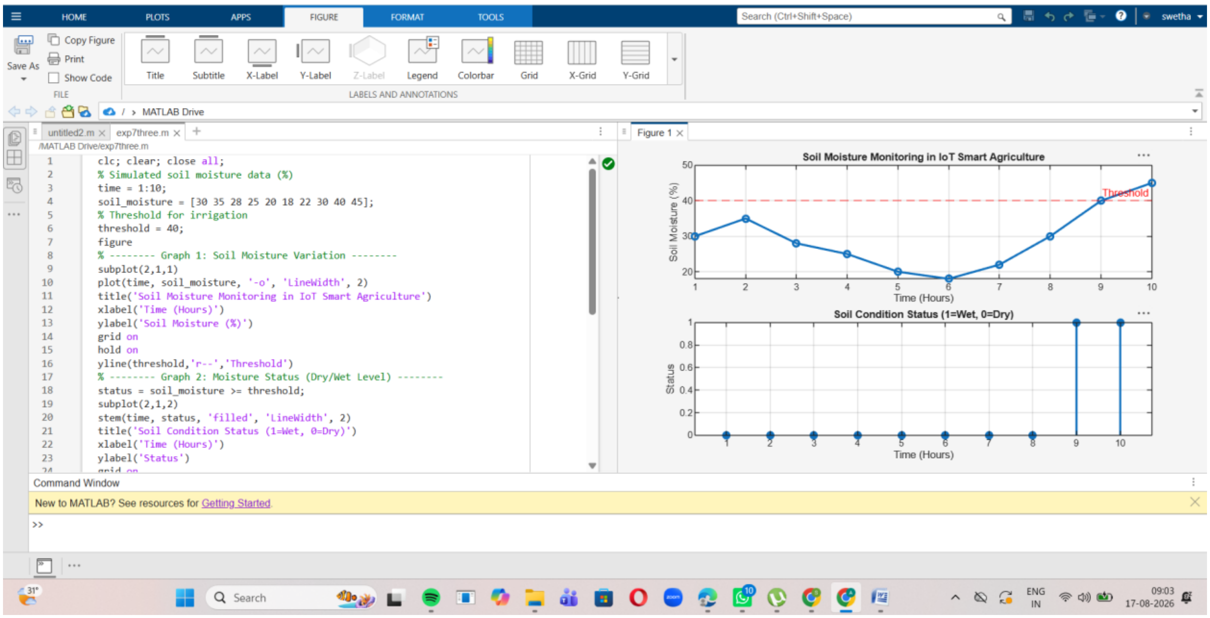
1. Define the agricultural parameters such as soil moisture, temperature, humidity, and water requirements for the crops.
2. Generate or collect sensor data representing the field conditions and import the data into MATLAB for processing.
3. Analyze the sensor values and set suitable threshold levels to determine whether irrigation or other actions are required.
4. Simulate the IoT-based control system in MATLAB to automatically control devices such as water pumps based on the sensor conditions.
5. Plot the sensor readings and system responses to evaluate the performance of the smart agriculture system under different field conditions.

objective 1 :

Program

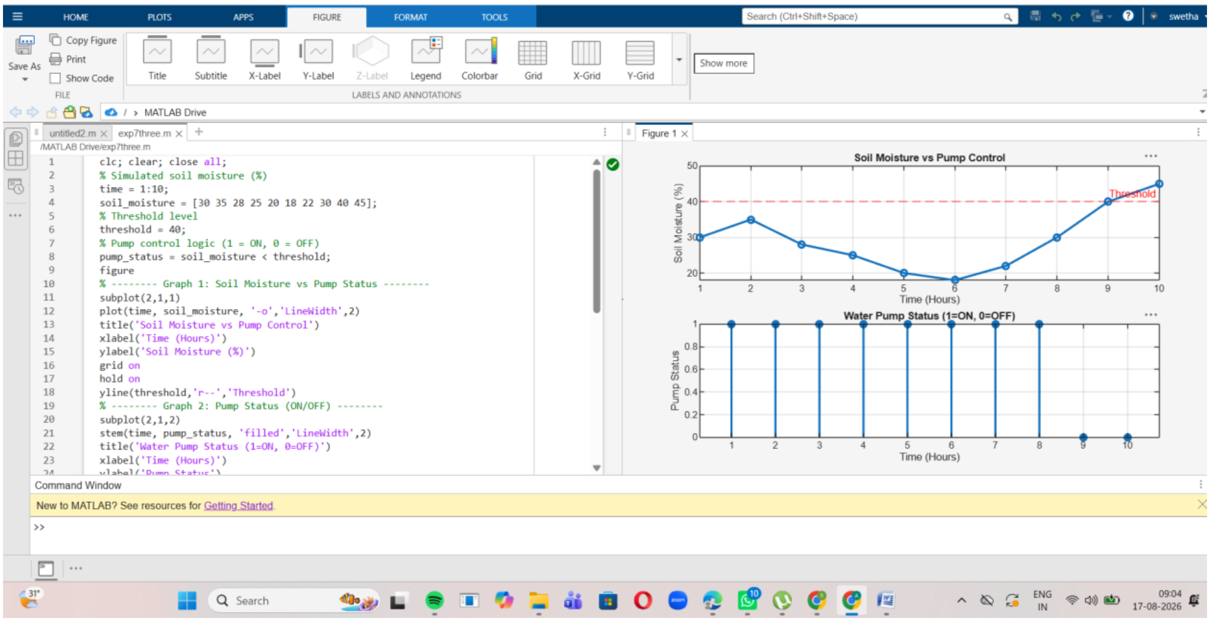
```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% -------- Graph 1: Soil Moisture Variation --------
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Moisture Status (Dry/Wet Level) --------
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on
```

output



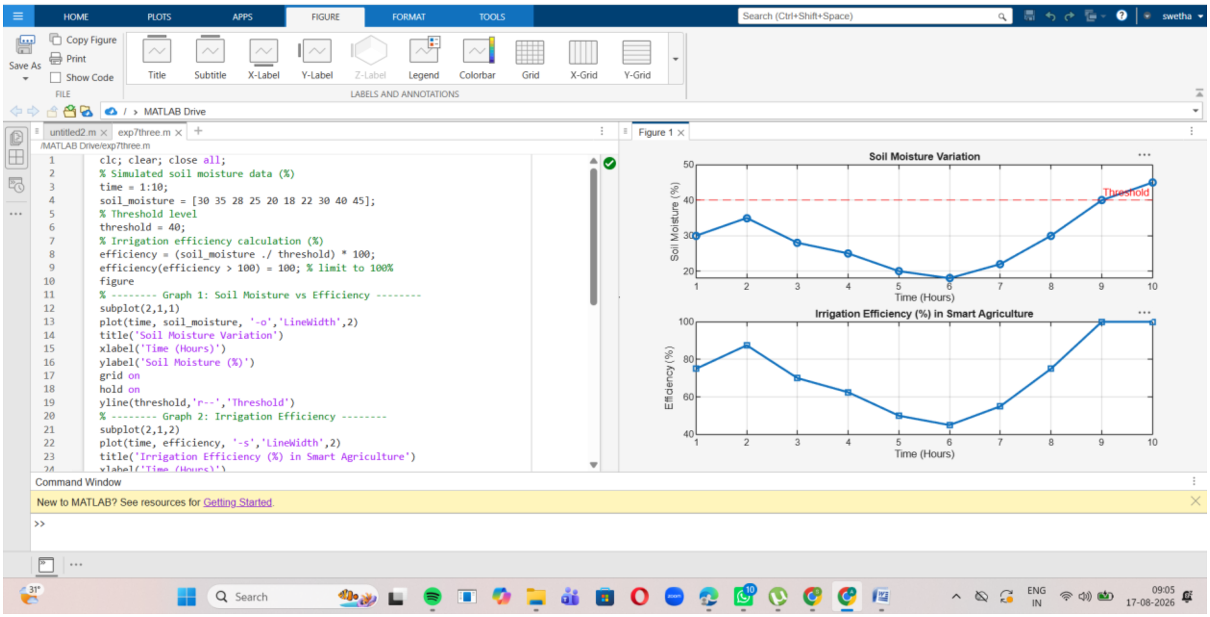
Objective 2: code and output:

```
clc; clear; close all;
% Simulated soil moisture (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;
figure
% -------- Graph 1: Soil Moisture vs Pump Status --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture vs Pump Control')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Pump Status (ON/OFF) --------
subplot(2,1,2)
stem(time, pump_status, 'filled','LineWidth',2)
title('Water Pump Status (1=ON, 0=OFF)')
xlabel('Time (Hours)')
ylabel('Pump Status')
grid on
```



Objective 3:code and output:

```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Irrigation efficiency calculation (%)
efficiency = (soil_moisture ./ threshold) * 100;
efficiency(efficiency > 100) = 100; % limit to 100%
figure
% -------- Graph 1: Soil Moisture vs Efficiency --------
subplot(2,1,1)
plot(time, soil_moisture, '-o','LineWidth',2)
title('Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% -------- Graph 2: Irrigation Efficiency --------
subplot(2,1,2)
plot(time, efficiency, '-s','LineWidth',2)
title('Irrigation Efficiency (%) in Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Efficiency (%)')
grid on
```



Result:

1.The soil moisture levels were successfully monitored using MATLAB simulation, showing variation over time and identifying dry and wet conditions based on threshold values.

2. The water pump control system was effectively implemented, where the pump automatically switched ON when soil moisture dropped below the threshold and remained OFF when moisture was sufficient.

3. The irrigation efficiency was successfully calculated and analyzed, showing that the system optimizes water usage by maintaining soil moisture close to the desired level.